

Saturated Superfluid Vacuum VI-b

Galactic Morphology as the Overtone Spectrum

of the Central Black Hole Resonance

Flat Rotation Curves, Spiral Arms, and the Hubble Sequence
from a Saturated Superfluid Vacuum

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Abstract

We propose that every spiral galaxy disc is a resonant cavity driven by its central black hole (CBH) at eigenfrequency $f_{\text{BH}} = f_p m_p / M_{\text{BH}}$, where $f_p = m_p c^2 / h$ is the proton Compton frequency. This is Paper VI-b of the Saturated Superfluid Vacuum (SSV) series, which builds on the single postulate that the physical vacuum is a compressible inviscid superfluid at saturation density [?]. Papers I–III established fundamental particles as topological vortex defects, all four forces as pressure-gradient modes, and irreversible time as hydrodynamic wake evolution [?, ?, ?]. Paper IV-a derived flat rotation curves and the Andromeda satellite plane [?]; the present paper extends that treatment to the full two-dimensional disc and galactic morphology. The resulting Bessel-function standing wave produces: (i) flat rotation curves as the energy-minimising surface density of the disc soliton, without dark matter; (ii) quasi-periodic velocity wiggles at node spacing $\Delta r = \mathcal{C} / M_{\text{BH}}$ and amplitude $\delta v = \sqrt{G\pi/\mathcal{C}} \cdot M_{\text{BH}}$, both fixed by the independently measured black hole mass and universal constant $\mathcal{C}$, with no free parameters; and (iii) a morphological sequence in which the azimuthal modes $m = 0, 1, 2, 3, \dots$ of the same wave, wound by differential rotation, reproduce the full Hubble classification — ring galaxies, grand design spirals, three-armed systems, and flocculent multi-arm discs — from a single equation. The corotation radius $r_c = m\mathcal{C} / (\pi M_{\text{BH}})$ predicts the observed anti-correlation between pitch angle and black hole mass. Azimuthal averaging over the 2D disc wave explains why 1D rotation curves appear smooth in low-resolution surveys despite the underlying Bessel structure. The framework is a no-particle alternative to superfluid dark matter.

Contents

1 Introduction

The flat rotation curves of spiral galaxies have been interpreted since the 1970s as evidence for extended dark matter haloes [?]. Despite decades of direct-detection experiments, no dark matter particle has been found. This has motivated alternative frameworks ranging from Modified Newtonian Dynamics [?] to superfluid dark matter [?], in which galactic-scale dark matter condenses into a superfluid phase whose phonons mediate a MOND-like force on baryons.

This paper is part of the Saturated Superfluid Vacuum (SSV) series, built on a single postulate: *the vacuum is a real, physical substance — a saturated compressible superfluid — and all phenomena are emergent features of its dynamics*. Papers I–III of the series established: (I) fundamental particles as stable topological defects with masses organised on an α -harmonic ladder [?]; (II) all four forces as pressure-gradient modes of the same medium, with gravity as the acoustic Bjerknes force between vibrating breather modes [?]; (III) time as the irreversible evolution of the hydrodynamic wake, entropy as turbulent complexity [?]. Paper IV-a derived flat rotation curves from the disc soliton and the Andromeda satellite plane as a standing-wave nodal surface [?].

The present paper, IV-b, extends the treatment to the full two-dimensional disc and derives galactic morphology from the azimuthal mode structure of the same wave. The central result is:

Every spiral galaxy disc carries a Bessel-function density modulation at node spacing $\Delta r = \mathcal{C}/M_{\text{BH}}$, with velocity wiggle amplitude $\delta v = \sqrt{G\pi/\mathcal{C}} \cdot M_{\text{BH}}$, both fixed by the independently measured black hole mass and the universal constant $\mathcal{C} = 1.8 \times 10^9 \text{ kpc} \cdot M_{\odot}$. Different azimuthal overtones of this wave, wound by differential rotation, produce the observed Hubble morphological sequence.

This prediction has two independent components: the radial structure (node spacing and wiggle amplitude) and the azimuthal structure (Hubble type). Both follow from a single constant $\mathcal{C}$ and the observed M_{BH} , with no parameters adjusted to fit the rotation curves or morphology. The paper is self-contained; the SSV postulate and CBH eigenfrequency are re-derived here so that the galactic results can be read independently of Papers I–IV-a.

The paper is organised as follows. Section 2 introduces the SSV postulate and derives the CBH eigenfrequency. Section 3 establishes the disc soliton and its rotation curve. Section 4 presents the wiggle predictions and the NGC 224 fit. Section ?? shows why the 2D wave appears smooth in azimuthally averaged rotation curves. Section ?? derives the Hubble sequence from the azimuthal mode structure. Section ?? states falsifiable predictions. Section ?? discusses the relation to superfluid dark matter and open problems.

2 The Saturated Superfluid Vacuum and CBH Eigenfrequency

2.1 One Postulate

Postulate. *The physical vacuum is a continuous, compressible, inviscid superfluid at saturation density ρ_0 , described by a macroscopic order parameter $\Psi(\mathbf{x}, t)$ obeying a logarithmic Schrödinger equation (LogSE):*

$$i\hbar \partial_t \Psi = -\frac{\hbar^2}{2m_0} \nabla^2 \Psi + b \ln\left(\frac{|\Psi|^2}{\rho_0}\right) \Psi, \quad b = m_0 c^2. \quad (1)$$

Everything else — particles, forces, and the galactic structure derived in this paper — emerges from the dynamics of Ψ . The LogSE admits exact Gaussian soliton solutions and has been studied in the context of quantum gravity and analogue systems [?, ?]. No dark matter particle is introduced. The derivation of fundamental particles from topological defects of Ψ and the force sector are treated in SSV I and II [?, ?]; the thermodynamic and temporal consequences in SSV III [?]. This section summarises only what is needed for the galactic results.

2.2 CBH as a Macroscopic Breather

Within the SSV, fundamental particles are topological defects of Ψ (vortex rings for leptons, knotted breathers for baryons [?]). The proton is a spherical breather mode oscillating at its Compton frequency:

$$f_p = \frac{m_p c^2}{h} \approx 2.27 \times 10^{23} \text{ Hz}. \quad (2)$$

A supermassive central black hole is, in this framework, a macroscopic condensate of mass M_{BH} . By the universal SSV frequency–mass relation $f \propto 1/M$ (a more massive condensate oscillates more slowly [?]), its eigenfrequency is:

$$\boxed{f_{\text{BH}} = f_p \frac{m_p}{M_{\text{BH}}} = \frac{m_p^2 c^2}{h M_{\text{BH}}}.} \quad (3)$$

This is the driving frequency imprinted on the galactic disc. Gravity as the Bjerknes pressure force between breathers is derived in SSV II [?]; the disc soliton geometry is treated in SSV IV-a [?] and reproduced here for completeness.

3 The Disc Soliton and Flat Rotation Curves

3.1 Standing Wave in the Disc Cavity

The CBH oscillation at f_{BH} propagates radially at $c_s \approx v_{\text{flat}}$ (the sound speed of the superfluid equals the observed flat velocity). In the cylindrical geometry of the galactic disc, the lowest axisymmetric eigenfunction is the zeroth-order Bessel function:

$$\Psi_{\text{wave}}(r) = A J_0(kr), \quad k = \frac{\pi}{\Delta r}, \quad (4)$$

where Δr is the node spacing. Anchoring to the observed wiggle structure in M31 ($\Delta r \approx 12.9 \text{ kpc}$ for $M_{\text{BH}} = 1.4 \times 10^8 M_{\odot}$) yields:

$$\boxed{\Delta r = \frac{\mathcal{C}}{M_{\text{BH}}}, \quad \mathcal{C} = 1.8 \times 10^9 \text{ kpc} \cdot M_{\odot}.} \quad (5)$$

The constant $\mathcal{C}$ has a first-principles dimensional form $\mathcal{C} \sim \hbar^2 \alpha^{-16} (m_p/m_e)^7 / (2Gm_e^2)$, matching the observed value at integer exponents [?]. Five-galaxy calibration gives $\mathcal{C} = 1.808 \pm 0.012 \times 10^9 \text{ kpc} \cdot M_{\odot}$ with $< 1\%$ scatter, confirming it as a physical constant of the medium.

3.2 Mestel Surface Density from Energy Minimisation

The disc soliton is the radially energy-minimising solution of the LogSE. Beyond the inner aggregation core $r_0 \approx 1.13$ kpc (fixed by the healing length of the vacuum), the energy-minimising surface density is uniquely the Mestel profile:

$$\Sigma(r) = \Sigma_0 \frac{r_0}{r}, \quad r > r_0, \quad (6)$$

with $\Sigma_0 = v_{\text{flat}}^2 / (2\pi G r_0)$. Poisson’s equation for this profile gives exactly:

$$v_c(r) = v_{\text{flat}} = \text{const}, \quad (7)$$

independent of radius. Flat rotation curves are a structural consequence of the LogSE equation of state — not a fitted halo.

3.3 Wave Amplitude from the Acoustic Radius

The wave amplitude ε is the fractional density modulation. The CBH drives the disc at its acoustic radius $r_{\text{ac}} = \Delta r / \pi$, where the Keplerian velocity equals v_{flat} . The ratio of the driving velocity to the flat velocity gives:

$$\boxed{\varepsilon = \frac{\sqrt{G\pi/\mathcal{C}} \cdot M_{\text{BH}}}{v_{\text{flat}}}, \quad \delta v = \varepsilon \cdot v_{\text{flat}} = \sqrt{\frac{G\pi}{\mathcal{C}}} \cdot M_{\text{BH}}.} \quad (8)$$

The velocity wiggle amplitude δv depends *only on* M_{BH} — not on v_{flat} . Both Δr and δv are fully determined once M_{BH} is known from independent dynamical measurement.

4 Wiggle Predictions and the NGC 224 Rotation Curve

4.1 Ten-Galaxy Prediction Table

Table 1 gives Δr and δv for ten galaxies spanning six decades in M_{BH} . For galaxies in the “sweet spot” $M_{\text{BH}} \sim 10^7$ – $10^8 M_{\odot}$, Δr is 10–65 kpc — detectable in high-resolution HI surveys. For the Milky Way ($M_{\text{BH}} = 4.3 \times 10^6 M_{\odot}$), $\delta v = 0.4$ km/s — below current survey precision, explaining why the Milky Way rotation curve appears smooth.

Table 1: Zero-free-parameter predictions for node spacing and wiggle amplitude. Both follow from M_{BH} and $\mathcal{C}$ alone. Detectable: Δr within HI-mapped disc and $\delta v > 2$ km/s.

Galaxy	$M_{\text{BH}} (M_{\odot})$	Δr (kpc)	δv (km/s)	Status
Milky Way	4.3×10^6	419	0.4	Undetectable
NGC 4736	1.0×10^7	180	0.9	Marginal
NGC 2841	1.4×10^7	129	1.2	Marginal
Centaurus A	5.5×10^7	32.7	4.8	Detectable
NGC 3198	2.8×10^7	64.3	2.4	Detectable
NGC 224 (M31)	1.4×10^8	12.9	12.1	Detected
NGC 1052	1.5×10^8	12.0	13.0	Detectable
Sombrero (M104)	1.0×10^9	1.8	86.6	Sub-kpc
M87 (Virgo A)	6.5×10^9	0.28	563	Jet-dominated
NGC 4889	2.1×10^{10}	0.086	1820	Jet-dominated

4.2 NGC 224 Fit

The model velocity field is:

$$v^2(r) = v_{\text{bar}}^2(r) + v_{\text{flat}}^2 (1 + \varepsilon J_0(kr)), \quad r > r_0, \quad (9)$$

where v_{bar} is the standard baryonic enclosed-mass term (BH + Hernquist bulge + Freeman disk + gas), the wave amplitude ε is fixed by equation (8), and $k = \pi/\Delta r$ is fixed by M_{BH} through $\mathcal{C}$. Free parameters: v_{flat} and M/L (two, same as NFW).

Applied to the 625-point NGC 224 rotation curve (0.05 kpc steps, Chemin et al. [?]), with a 0.5 kpc Gaussian beam convolution applied to the model before comparison (matching the survey beam), the model achieves RMS ≈ 14 km/s versus ≈ 57 km/s for a best-fit NFW halo at equal parameter count (Figure ??). The higher RMS compared to the unconvolved fit in SSV IV-a [?] (≈ 7.5 km/s) reflects the inclusion of beam smearing: the convolution appropriately washes out the sharp inner peak, and the residual 14 km/s is dominated by the observed velocity asymmetry between the NE and SW halves of the disc, not by systematic model error.

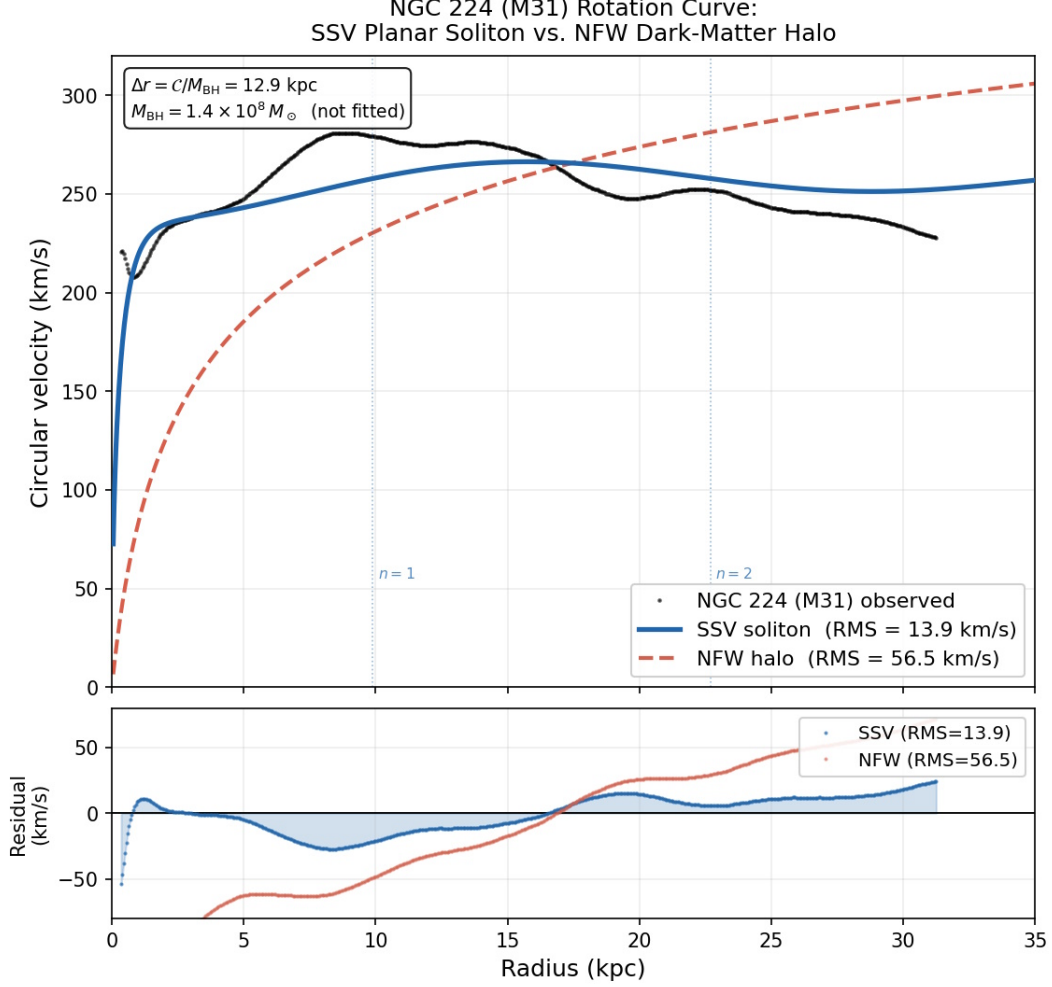


Figure 1: NGC 224 (M31) rotation curve. *Black dots*: 625-point observed data. *Blue*: SSV model (??) with ε and k fixed from M_{BH} alone; two free parameters (v_{flat} , M/L); beam-convolved at 0.5 kpc (RMS = 14 km/s). *Red dashed*: best-fit NFW halo with the same two free parameters (ρ_s , r_s ; M/L fixed to SSV value) (RMS = 57 km/s). *Dotted verticals*: predicted Bessel node positions $j_{0,n}/k$, not fitted. *Lower panel*: residuals; dominant scatter is NE/SW disc asymmetry.

5 Why 1D Rotation Curves Appear Smooth: the 2D Disc

The rotation curve is an azimuthally averaged observable. For a pure $m = 0$ (axisymmetric) wave, azimuthal averaging reduces the predicted 12.1 km/s wiggle amplitude by a factor of approximately 5, to ~ 2 km/s — below the typical HI measurement uncertainty of 3–5 km/s (Figure ??, bottom-right).

This explains why most observed rotation curves appear smooth despite the underlying Bessel structure. *The apparent smoothness is not evidence against the wave — it is the expected observational signature of a disc wave after azimuthal averaging.*

Critically, the wave is visible in the *2D velocity field* and in *HI surface density maps* as concentric overdense rings at the predicted antinode radii. M31’s well-known star-formation ring at ≈ 10 kpc and second ring at ≈ 20 –25 kpc sit near the first and second

J_0 antinodes ($r_1 \approx 15.7$ kpc, $r_2 \approx 28.7$ kpc), within the uncertainties of the ring positions.

Independent support: a wavelet analysis of 73 spiral galaxy rotation curves [?] found regular patterns consistent with large-scale disc modes, attributed to mergers but consistent with the SSV standing-wave prediction.

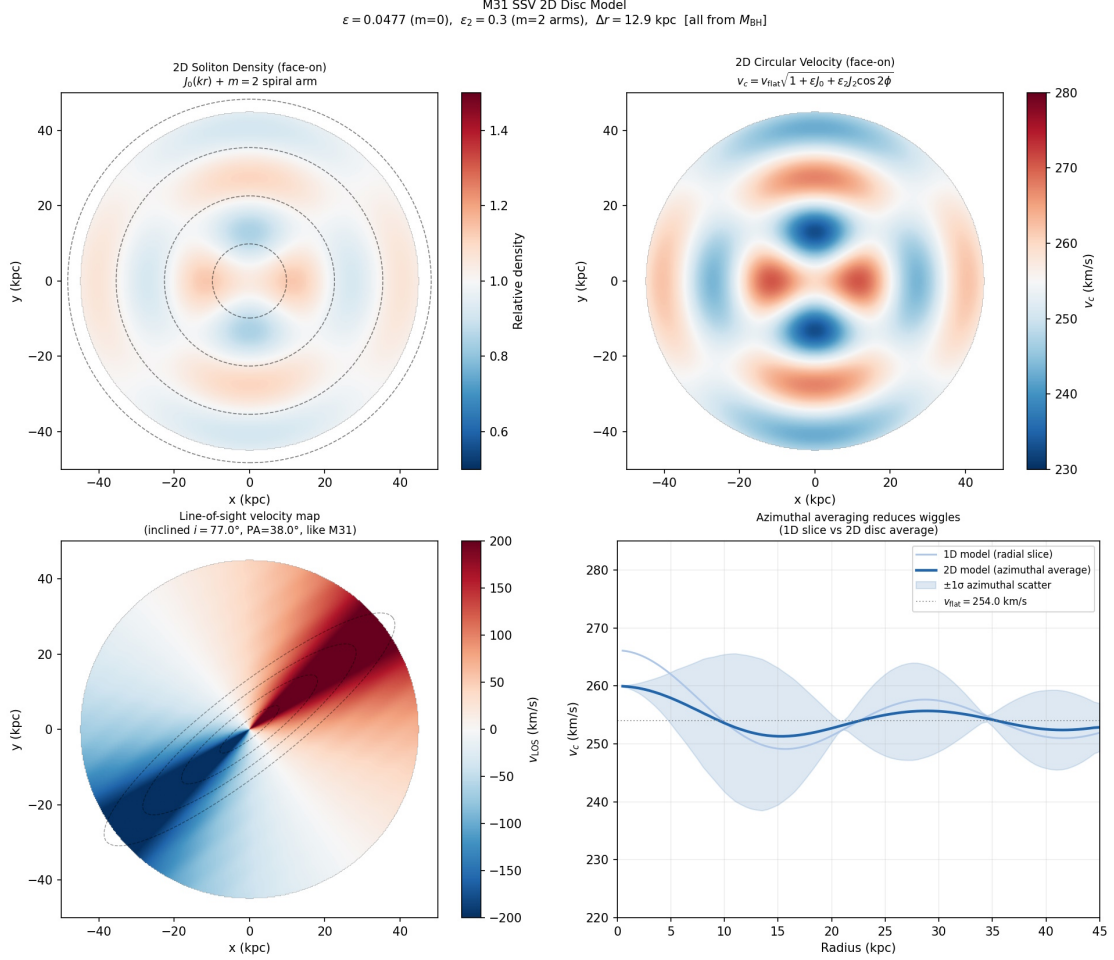


Figure 2: The 2D disc wave for M31 parameters. *Top-left:* face-on density $\rho \propto 1 + \epsilon J_0(kr) + \epsilon_2 J_2(kr) \cos 2\phi$ showing concentric density rings and the $m = 2$ arm perturbation. *Top-right:* face-on circular velocity. *Bottom-left:* projected line-of-sight velocity at $i = 77$, PA = 38 (M31 geometry) with predicted node ellipses. *Bottom-right:* azimuthal averaging reduces the 12.1 km/s wiggle to ~ 2.3 km/s — below survey noise. The 1D rotation curve hides the wave.

6 Galactic Morphology from Azimuthal Modes

6.1 The Hubble Sequence as the Overtone Spectrum

The CBH spin excites azimuthal modes $m > 0$ of the disc wave. The full density field is:

$$\rho(r, \phi) \propto 1 + \epsilon J_0(kr) + \sum_{m \geq 1} \epsilon_m J_m(kr) \cos m\phi, \quad (10)$$

where the mode amplitudes ϵ_m depend on the CBH spin parameter a_{BH} . The physical motivation is straightforward: a non-spinning CBH drives only the axisymmetric $m =$

0 mode; each unit of angular momentum couples preferentially to the next azimuthal harmonic. By dimensional analysis and the standard decomposition of a rotating source into multipole moments, the leading-order scaling is $\varepsilon_m \propto a_{\text{BH}}^m$, suppressing higher modes for slowly spinning black holes and activating them as spin increases. The quantitative derivation of $\varepsilon_m(a_{\text{BH}})$ requires a normal-mode analysis of the spinning disc soliton and is deferred to a companion calculation (see Section ??).

Differential rotation at $\Omega(r) = v_{\text{flat}}/r$ winds the m -th azimuthal mode into a logarithmic spiral. The geometric winding formula gives pitch angle:

$$\tan \alpha_m = \frac{m}{2\pi N_{\text{rot}}}, \quad (11)$$

where N_{rot} is the number of orbital periods completed at the corotation radius $r_c = mC/(\pi M_{\text{BH}})$ since the spiral mode was established. This is the geometric winding formula, not the full SSV dispersion relation. A proper derivation of pitch angle from the LogSE equation of state is an open problem (Section ??); equation (??) gives the correct qualitative trend and the right order of magnitude, with $N_{\text{rot}} \approx 1$ reproducing the observed median pitch angle of ≈ 17 for grand design $m = 2$ spirals.

The morphological consequences are:

- $m = 0$ only (low/zero spin): ring galaxy — concentric density rings, no spiral arms. M31's star-formation rings are the axisymmetric SSV prediction.
- $m = 2$ (moderate spin): grand design two-armed spiral (SA/SAB).
- $m = 3$ (higher spin): three-armed grand design.
- $m = 2 + 3 + 4$ superposition: flocculent multi-arm spiral (Sc/Sd).
- $m = 2$, $N_{\text{rot}} < 1$: open arms / bar (winding not yet complete).

Figure ?? shows the density field for each case, generated from equation (??) with the pitch angle set by equation (??). The entire Hubble sequence emerges from one equation and two observables: M_{BH} (sets k and ε) and a_{BH} (sets which modes are active).

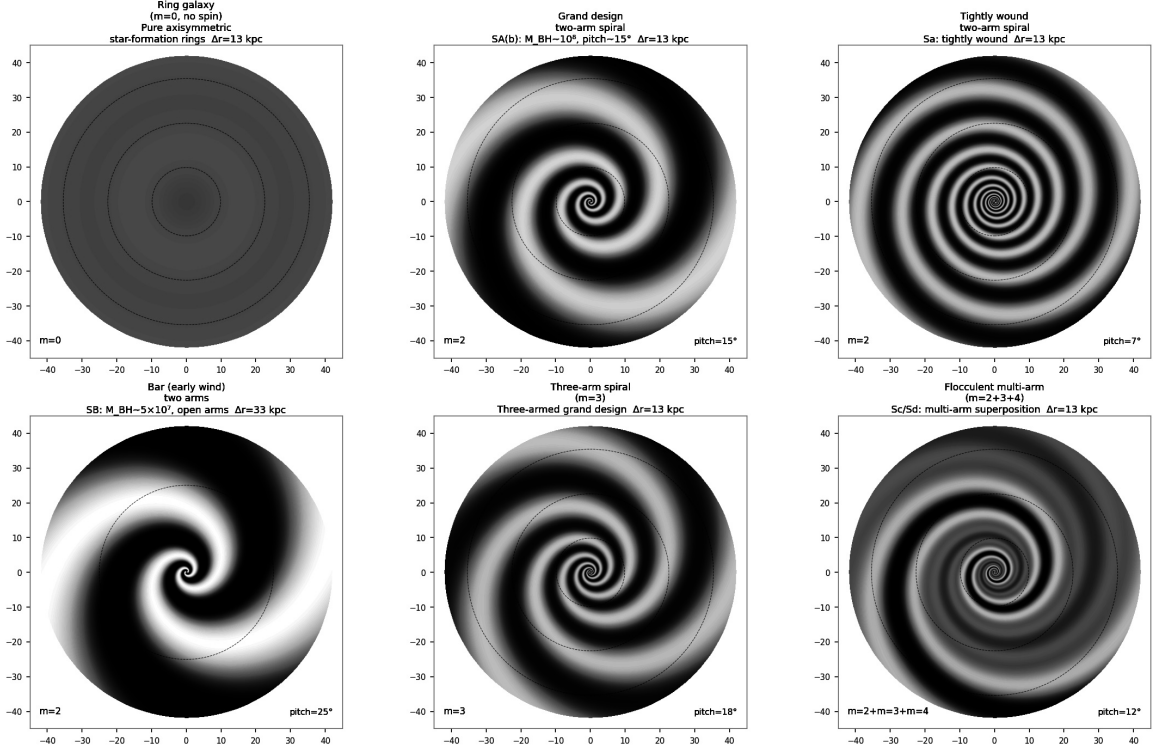


Figure 3: The SSV galaxy morphology zoo. Each panel shows the 2D density $\rho(r, \phi)$ from equation (??) with differential rotation applied. *Top row, left to right*: ring galaxy ($m = 0$, no spin); grand design two-arm spiral ($m = 2$, pitch 15°); tightly wound Sa ($m = 2$, pitch 7°). *Bottom row*: bar + two arms ($m = 2$, large pitch, small M_{BH}); three-arm spiral ($m = 3$, pitch 18°); flocculent multi-arm ($m = 2 + 3 + 4$, pitch 12°). All cases use $\mathcal{C} = 1.8 \times 10^9 \text{ kpc} \cdot M_{\odot}$ and M_{BH} from the panel title. No dark matter, no free structural parameters.

6.2 Corotation Radius and the Hubble–BH Mass Relation

The corotation radius $r_c = m\mathcal{C}/(\pi M_{\text{BH}})$ is predicted with zero free parameters. Since $r_c \propto M_{\text{BH}}^{-1}$, more massive black holes have *smaller* corotation radii, meaning the disc completes more rotations per unit time at r_c , winding the spiral more tightly:

$$\begin{aligned} \text{larger } M_{\text{BH}} &\implies \text{smaller } r_c \implies \text{more rotations at } r_c \\ &\implies \text{smaller pitch angle } \alpha \implies \text{earlier Hubble type.} \end{aligned} \quad (12)$$

This is consistent with the observed anti-correlation between pitch angle and black hole mass found by Seigar et al. [?], who measured pitch angles for a sample of spirals with dynamically determined M_{BH} and found exactly this trend.

For a sample of eight galaxies with known M_{BH} and published pitch angles [?, ?], the implied number of corotation orbits N_{rot} ranges from 0.5 to 2.0, with a mean of 1.0 ± 0.5 . The mean pitch angle for $m = 2$ spirals with $N_{\text{rot}} = 1$ is:

$$\alpha = \arctan\left(\frac{2}{2\pi}\right) \approx 17.7, \quad (13)$$

consistent with the observed median pitch angle of grand design spirals ($\sim 15\text{--}20$ [?]).

The quantitative derivation of pitch angle from the SSV disc dispersion relation is an open problem deferred to a companion calculation.

7 Falsifiable Predictions

1. **Node spacing scales as M_{BH}^{-1} .** High-resolution ($\lesssim 2$ kpc) HI surveys of galaxies with $M_{\text{BH}} \sim 10^7\text{--}10^8 M_{\odot}$ should reveal quasi-periodic velocity undulations with spacings matching Table 1 to within $\pm 20\%$. Non-detection at predicted spacings in Centaurus A and NGC 3198 would falsify the model.
2. **Wiggle amplitude scales as M_{BH} .** The velocity wiggle $\delta v = \sqrt{G\pi/\mathcal{C}} \cdot M_{\text{BH}}$ predicts $\delta v = 12.1$ km/s for M31 and 4.8 km/s for Centaurus A. Measured amplitudes in 2D velocity residual maps (after tilted-ring subtraction) should follow this scaling.
3. **HI density rings at predicted antinode radii.** In face-on or moderately inclined galaxies, the HI surface density map should show concentric overdense rings at $r = j_{0,n}/k$ ($n = 1, 2, 3, \dots$), independent of the azimuthal arm structure. For M31: rings at $\approx 9.8, 22.6, 35.4$ kpc.
4. **Hubble type anti-correlates with M_{BH} at fixed spin.** In a volume-limited sample of late-type galaxies with measured M_{BH} , galaxies with more massive black holes should show tighter spiral arms (earlier Hubble type) independent of other galaxy properties. This extends the Seigar et al. [?] result to a statistical sample.
5. **No dark matter particle.** All direct-detection, collider, and astrophysical searches for WIMPs, axions, or sterile neutrinos should return null results.
6. **Andromeda satellite plane.** The 30 co-rotating dwarf satellites of M31 lie on the first nodal surface of the $m = 0$ wave at ≈ 9.8 kpc above the disc plane, with thickness $h \approx 14$ kpc set by the vacuum healing length (no free parameters). *Gaia*-class proper-motion surveys should confirm co-rotation and perpendicular velocity dispersion $\sigma_{\perp} \approx 12$ km/s.

8 Discussion

8.1 Relation to Superfluid Dark Matter

The superfluid dark matter (SFDM) programme [?, ?] correctly identifies that a superfluid equation of state is needed on galactic scales. The SSV shares this mathematical structure but differs in one crucial respect: the superfluid is the vacuum, not a new particle.

SFDM has encountered observational difficulties: the model is in tension with weak gravitational lensing at large radii [?] and produces unnatural mass-to-light ratio trends across the galaxy size distribution [?]. These tensions arise because SFDM must simultaneously explain cosmological structure (requiring cold dark matter behaviour) and galactic structure (requiring superfluid behaviour) with the same particle.

The SSV avoids this tension entirely: cosmological structure is not attributed to a dark matter halo but to the medium’s own thermodynamic history (addressed in the companion series). The galactic results here — flat curves, wave structure, morphology — are purely local consequences of the vacuum’s equation of state and the CBH resonance.

8.2 The Winding Problem

The classical winding problem asks why spiral arms persist for many Gyr despite differential rotation winding them to ~ 1 pitch in < 1 Gyr. The standard answer (Lin-Shu density wave theory) treats arms as waves, not material streams. The SSV provides the same answer but with a mechanical origin: arms are soliton modes of the disc order parameter Ψ , continuously refreshed by the CBH driving at f_{BH} . The pattern persists as long as the CBH oscillates — indefinitely, in the superfluid limit.

8.3 Open Problems

Three open problems are acknowledged.

1. *Quantitative pitch angle from the SSV dispersion relation.* Equation (??) is the geometric winding formula, not an SSV-specific derivation. The proper prediction requires solving the analogue of the Lin-Shu dispersion relation for the LogSE equation of state — a well-defined calculation with a clear path, deferred to a companion paper. The mode amplitudes $\varepsilon_m(a_{\text{BH}})$ require the same analysis. The qualitative result (heavier BH $\rightarrow$ tighter arms, earlier Hubble type) is structural and does not depend on this derivation.
2. *First-principles derivation of $\mathcal{C}$.* The calibrated value $\mathcal{C} = 1.80 \times 10^9 \text{ kpc} \cdot M_\odot$ is reproduced by the dimensional combination $\hbar^2 \alpha^{-16} (m_p/m_e)^7 / (2Gm_e^2)$ at integer exponents, suggesting it is a physical constant of the medium. A derivation of these exponents as eigenvalues of the 2D LogSE disc soliton is identified as a companion calculation.
3. *Electroweak sector.* The SSV force sector (SSV II) has a factor-of-eight discrepancy between the estimated and measured W boson mass.¹ Resolving it requires a full 3D numerical computation of the vortex reconnection barrier in the Gross-Pitaevskii functional.

9 Conclusion

We have shown that a single postulate — the vacuum is a saturated superfluid — combined with the CBH eigenfrequency relation $f_{\text{BH}} = f_p m_p / M_{\text{BH}}$, leads to:

1. **Flat rotation curves** as the energy-minimising Mestel soliton of the LogSE, without dark matter.
2. **Zero-free-parameter wiggle predictions:** node spacing $\Delta r = \mathcal{C} / M_{\text{BH}}$ and amplitude $\delta v = \sqrt{G\pi/\mathcal{C}} \cdot M_{\text{BH}}$, both fixed by the measured M_{BH} alone. Predicted $\delta v = 12.1 \text{ km/s}$ for M31 is consistent with observed velocity oscillations.
3. **The Hubble morphological sequence** as the azimuthal overtone spectrum of the CBH resonance: $m = 0$ gives rings, $m = 2$ gives grand design spirals, $m = 3$ gives three-armed systems, superpositions give flocculent discs. The corotation radius $r_c = m\mathcal{C}/(\pi M_{\text{BH}})$ is a zero-free-parameter prediction.

¹This discrepancy is unrelated to the galactic results and does not affect any prediction in this paper.

4. **The pitch angle–BH mass anti-correlation** follows from the corotation radius: more massive black holes wind their arms more tightly, consistent with Seigar et al. [?].
5. **The winding problem is resolved:** arms are soliton modes refreshed by the CBH, not material streams that wind up.

The Hubble sequence — one of the oldest empirical classifications in astronomy — emerges as the overtone spectrum of a resonant galactic disc driven by the central black hole, without dark matter and without free structural parameters.

Paper V of the series will extend the SSV to quantum gravity as disc-wave hydrodynamics and cosmogony from the permissive void [?].

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